Ring Homomorphism $\pi : R \longrightarrow R/I$

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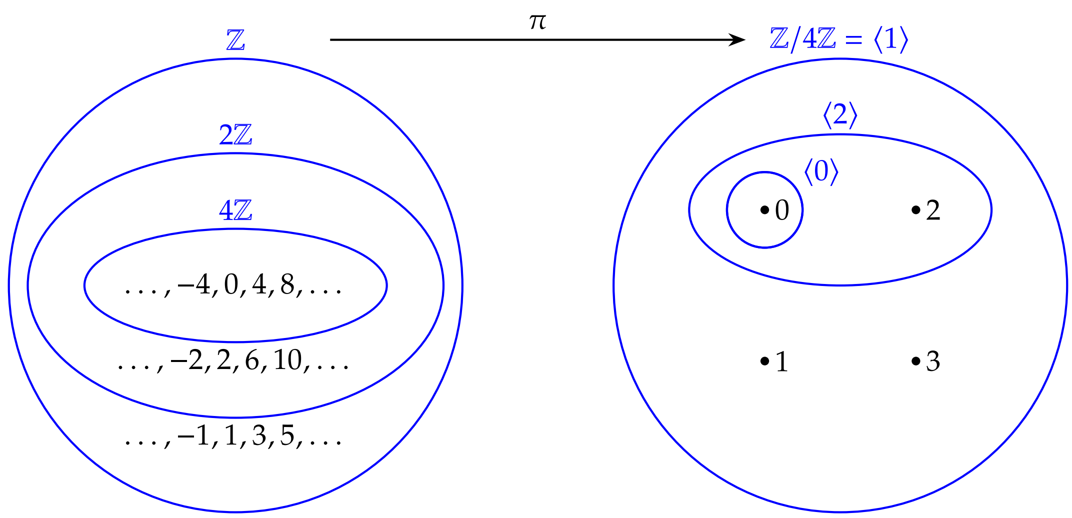
Observation. In the ring of integers $\mathbb{Z}$, there are three ideals containing $4\mathbb{Z}$:

$$4\mathbb{Z} \subset 2\mathbb{Z} \subset \mathbb{Z}.$$

In the ring $\mathbb{Z}/4\mathbb{Z}$, there are three ideals total:

$$\begin{aligned}\langle 0\rangle &= \left\{0\cdot x : x \in \mathbb{Z}/4\mathbb{Z}\right\} = \{0\}\,,\\ \langle 1\rangle &= \left\{1\cdot x : x \in \mathbb{Z}/4\mathbb{Z}\right\} = \{0,1,2,3\} = \langle 3\rangle \text{ and}\\ \langle 2\rangle &= \left\{2\cdot x : x \in \mathbb{Z}/4\mathbb{Z}\right\} = \{0,2\}\,.\end{aligned}$$



Here, there is a bijection $\{4\mathbb{Z}, 2\mathbb{Z}, \mathbb{Z}\} \longrightarrow \left\{\langle 0\rangle, \langle 1\rangle, \langle 2\rangle\right\}$.

- Let R be a ring, and $I \subseteq R$ be an ideal.
- We writes "$I \lhd R$" to indicate that I is an ideal of R.
- The natural projection map is

$$\begin{array}{rccc} \pi & : & R & \longrightarrow & R/I \\ & & r & \longmapsto & r+I \end{array}$$

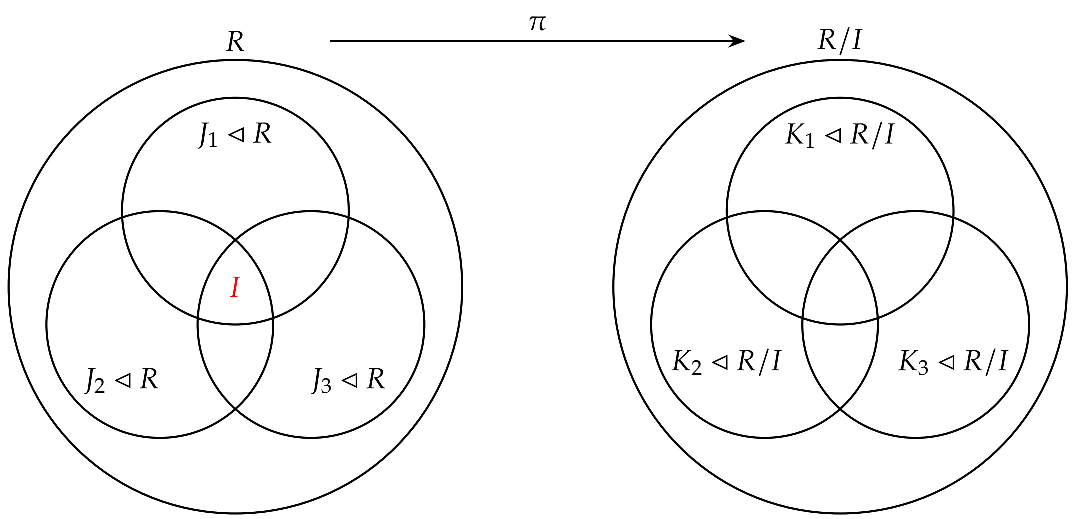
  for all $r \in R$.
- Let

$$\mathcal{L}_I(R) := \{J \subseteq R : I \subseteq J \text{ and } J \lhd R\}$$

  is the set of ideals of R that contain I, and let

$$\mathcal{L}(R/I) := \left\{K \subseteq R/I : K \lhd R/I\right\}$$

  is the set of ideals of the quotient ring R/I.



Theorem. *There is a one-to-one correspondence between the set of ideals of R containing I and the set of ideals of R/I. That is, there is a bijection*

$$\phi : \mathcal{L}_I(R) \to \mathcal{L}(R/I).$$

Proof. Consider

$$\begin{array}{rccc} \phi & : & \mathcal{L}_I(R) & \longrightarrow & \mathcal{L}(R/I) \\ & & J & \longmapsto & J/I \end{array}$$

and

$$\begin{array}{rcccl} \psi & : & \mathcal{L}(R/I) & \longrightarrow & \mathcal{L}_I(R) \\ & & K & \longmapsto & \pi^{-1}[K] = \{r \in R : r + I \in K\} \end{array}.$$

We must show that

- $\psi[\phi[J]] = J$ for all $J \subseteq R$ s.t. $I \subseteq J$;
- $\phi[\psi[K]] = K$ for all $K \subseteq R/I$.

(i) Let $J \in \mathcal{L}_I(R)$, i.e., $I \subseteq J \subseteq R$ and $J \lhd R$. Then

$$\phi[J] = J/I = \{r + I : r \in J\},$$

and so

$$\begin{aligned} \psi[\phi[J]] &= \psi[J/I] \\ &= \left\{r \in R : r + I \in J/I\right\}, \\ &= \{r \in R : r \in J\}, \\ &= J. \end{aligned}$$

(ii) Let $K \in \mathcal{L}(R/I)$, i.e., $K \subseteq R/I$ and $K \lhd R/I$. Then

$$\psi[K] = \pi^{-1}[K] = \{r \in R : r + I \in K\},$$

and so

$$\begin{aligned} \phi[\psi[K]] &= \phi[\pi^{-1}[K]] \\ &= \left\{r + I : r + \pi^{-1}[K]\right\}, \\ &= \{r + I : r + I \in K\}, \\ &= K. \end{aligned}$$

$\square$